Analyse IV - Summary

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1 Information

A concise and rigorous summary transcribing the most important concepts and theorems from the Analyse IV lecture at EPFL.

2 Holomorphic and Analytic Functions

2.1 Definitions

Let $\Omega \subseteq \mathbb{C}$ be an open set. A complex function $f : \Omega \to \mathbb{C}$ is said to be **complex differentiable** at a point $z_0 \in \Omega$ if the following limit exists:

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

If f is complex differentiable at *every* point of Ω, we say f is **holomorphic** on Ω.
If f is holomorphic on the entire complex plane ($\Omega = \mathbb{C}$), we call f an **entire function** (e.g., polynomials, e^z, $\sin(z)$).

3 The Cauchy-Riemann Equations

Let $z = x + iy \in \mathbb{C}$. A complex function $f(z)$ can be separated into its real and imaginary parts:

$$f(x + iy) = u(x, y) + iv(x, y)$$

where $u, v : \mathbb{R}^2 \to \mathbb{R}$.

3.1 Necessary and Sufficient Conditions

Theorem: Suppose $u(x, y)$ and $v(x, y)$ have continuous first-order partial derivatives (i.e., they are of class C^1) on an open set Ω. The function $f = u + iv$ is holomorphic on Ω if and only if it satisfies the **Cauchy-Riemann equations**:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

When these equations hold, the complex derivative can be computed as:

$$f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} - i\frac{\partial u}{\partial y}$$

3.2 Harmonic Conjugates

Because holomorphic functions are infinitely differentiable, u and v have continuous second-order derivatives. By applying the Cauchy-Riemann equations and Schwarz's theorem (symmetry of second derivatives), we get:

$$\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \text{and} \quad \Delta v = \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$$

Therefore, the real and imaginary parts of a holomorphic function are **harmonic functions**. We say that v is the harmonic conjugate of u.

4 The Complex Logarithm

4.1 Principal Argument

For any $z \neq 0$, the principal argument $\text{Arg}(z)$ is the unique angle in the interval $(-\pi, \pi]$. It is computed using the arctangent function depending on the quadrant of (x, y).
Crucial limitation: $\text{Arg}(z)$ is discontinuous across the negative real axis.
this is how we compute it :

$$\text{Arg}(z) = \begin{cases} \arctan\left(\frac{y}{x}\right) & \text{if } x > 0 \\ \arctan\left(\frac{y}{x}\right) + \pi & \text{if } x < 0 \text{ and } y \geq 0 \\ \arctan\left(\frac{y}{x}\right) - \pi & \text{if } x < 0 \text{ and } y < 0 \\ \frac{\pi}{2} & \text{if } x = 0 \text{ and } y > 0 \\ -\frac{\pi}{2} & \text{if } x = 0 \text{ and } y < 0 \end{cases}$$

4.2 Principal Branch of the Logarithm

We define the **principal branch** of the complex logarithm, denoted $\text{Log}(z)$, as:

$$\text{Log}(z) = \ln|z| + i\text{Arg}(z) = \ln\left(\sqrt{x^2+y^2}\right) + i\text{Arg}(x+iy)$$

Rigorous Properties:

- **Domain:** $\text{Log}(z)$ is a well-defined, single-valued function on the cut plane $\mathbb{C} \setminus (-\infty, 0]$. We must remove zero and the negative real axis (the branch cut) to maintain continuity.
- **Holomorphy:** On this cut plane, $\text{Log}(z)$ is holomorphic.
- **Derivative:** Its complex derivative is exactly what we expect from real analysis:

$$\frac{d}{dz}\text{Log}(z) = \frac{1}{z} \quad \text{for } z \in \mathbb{C} \setminus (-\infty, 0]$$

5 Complex Integration

Definition (Line Integral)
Let $\gamma : [a, b] \to \mathbb{C}$ be a regular (smooth) curve that parametrizes the path $\Gamma \subseteq \mathbb{C}$. Let $f : \Gamma \to \mathbb{C}$ be a continuous function. We define the line integral of f along Γ by:

$$\int_\Gamma f(z)\, dz = \int_a^b f(\gamma(t))\gamma'(t)\, dt$$

Example
Let Γ be the unit semi-circle centered at $z = 0$, parametrized counterclockwise: $\gamma(t) = e^{it}$ for $t \in [0, \pi]$. Then $dz = ie^{it}dt$.

5.1 Cauchy's Theorem

Let $U \subseteq \mathbb{C}$ be an open and simply connected set, and let γ be a closed and piecewise regular curve inside U. If $f : U \to \mathbb{C}$ is holomorphic, then:

$$\oint_\Gamma f(z)\, dz = 0$$

- **Simply connected:** The set consists of "one piece" and has no holes.
- **Piecewise regular:** Γ consists of finitely many smooth curves joined end-to-end (e.g., the boundary of a square).

5.1.1 Example

Let Γ be the unit circle and $f(z) = z^2$. Since f is holomorphic on the simply connected set $U = \mathbb{C}$, by Cauchy's Theorem:

$$\oint_\Gamma z^2\, dz = 0$$

5.2 The Extension of Cauchy's Theorem (Deformation)

Let Γ_0, Γ_1 be closed, simple, regular curves such that $\Gamma_1 \subseteq \operatorname{int}(\Gamma_0)$. Let γ_0, γ_1 be parametrizations of Γ_0, Γ_1 with the same orientation. If f is holomorphic on the region between Γ_0 and Γ_1 (including the boundaries), then:

$$\oint_{\Gamma_0} f(z)\, dz = \oint_{\Gamma_1} f(z)\, dz$$

5.3 Cauchy's Integral Formula

Suppose $D \subseteq \mathbb{C}$ is open and $f : D \to \mathbb{C}$ is holomorphic. If $z_0 \in D$, then:

$$f(z_0) = \frac{1}{2\pi i} \oint_\gamma \frac{f(z)}{z - z_0}\, dz$$

for every simple, closed, counterclockwise-oriented curve γ in D whose interior contains z_0.